Section B: Answer ALL questions. Parts marked with * involve knowledge of the extension component. Write your answers in the spaces provided. Take $g = 9.81$ m s^{-2}.

1. An empty glass cup is placed on a table under room temperature of 23 °C. 400 g of water at 100 °C is poured into the cup. After a moment, the temperature of water in the cup is found to be 93 °C.

(a) State a cause for the drop of water temperature other than heat loss to the surrounding air. (1 mark)

(b) Estimate the heat capacity H, in J °C^{-1}, of the glass cup. (2 marks)
Given: specific heat capacity of water = 4200 J kg^{-1} °C^{-1}

(c) (i) What is the kind of electromagnetic waves, other than visible light, mainly emitted from this cup of hot water ? (1 mark)

(ii) Name a device in daily life which makes use of this kind of electromagnetic waves. (1 mark)

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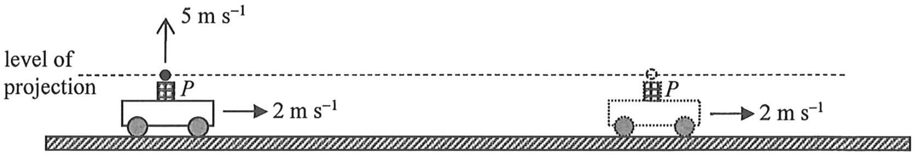
2. A trolley is moving on a horizontal surface with a constant velocity of 2 m s^{-1}. At time $t = 0$ s, a small bead of mass 0.01 kg is projected upwards with a vertical speed of 5 m s^{-1} from a device P mounted on the trolley as shown. Neglect air resistance.

Figure 2.1



*(a) Find the time taken, t, for the bead to travel back to the level of projection. (2 marks)

(b) What is the kinetic energy of the bead at its maximum height ? (2 marks)

*(c) When the bead falls back to the level of projection, it is just above device P as shown. Explain why the bead does not fall behind or beyond device P at that moment. (2 marks)

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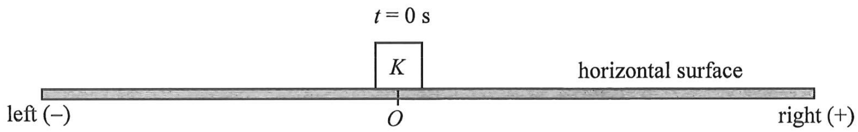
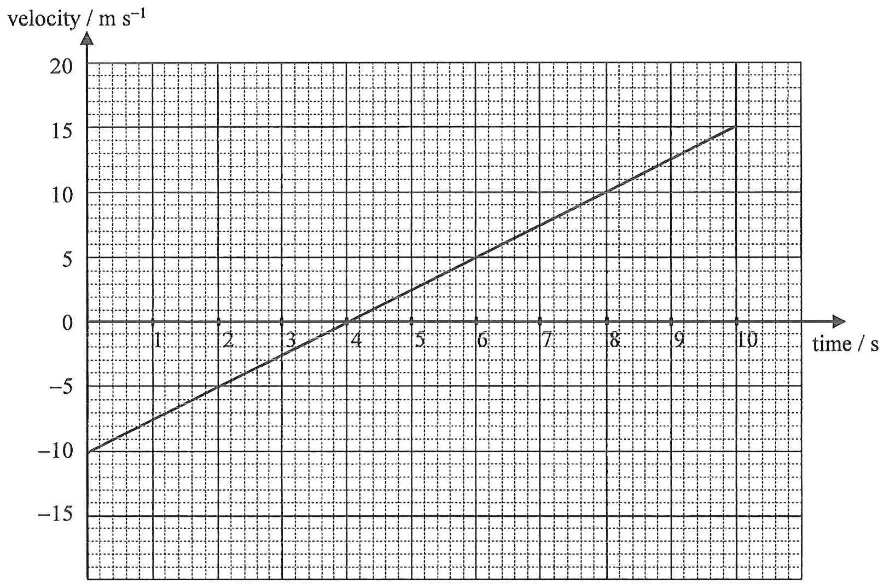
Figure 3.1



An object K moving on a horizontal surface is acted upon by a horizontal constant net force (not drawn on the diagram) such that it passes point O at time $t = 0$ s as shown in Figure 3.1. Its velocity-time graph is shown below, taking to the right as positive (+).



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(a) Briefly describe the motion of the object from $t = 0$ s to $t = 4$ s. (2 marks)

(b) Find the position of the object at $t = 10$ s. (2 marks)

(c) Find the magnitude of the net force acting on the object at $t = 4$ s. The mass of the object is 0.4 kg. (2 marks)

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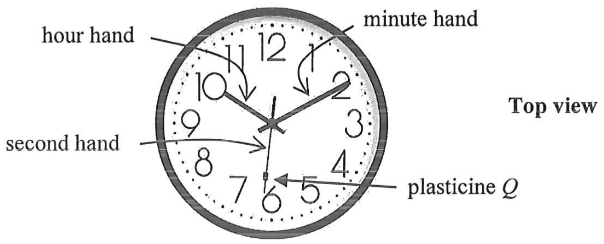
*4. A large clock facing upwards is placed horizontally on a table as shown. A small piece of plasticine Q rests on the steadily rotating second hand of the clock.

Figure 4.1



(a) Find the angular speed ω, in rad s^{-1}, of the second hand of the clock. (1 mark)

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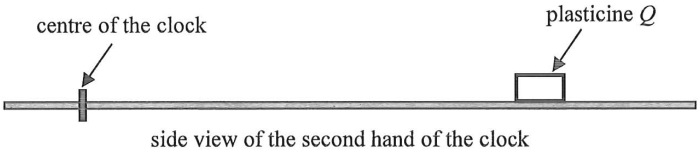
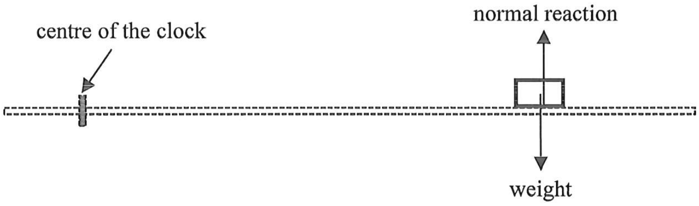
(b) Figure 4.2 shows the side view of the second hand of the clock.

Figure 4.2



In the free-body diagram of plasticine Q below, add a force acting on Q to complete the diagram and name this force. (1 mark)



(c) A student claims that the centripetal force required for Q will become smaller if it is placed nearer to the centre of the clock. Explain briefly whether this claim is correct or not. (2 marks)

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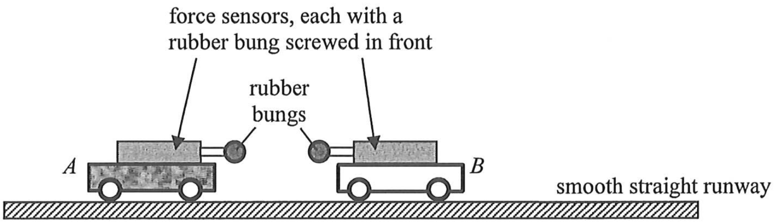
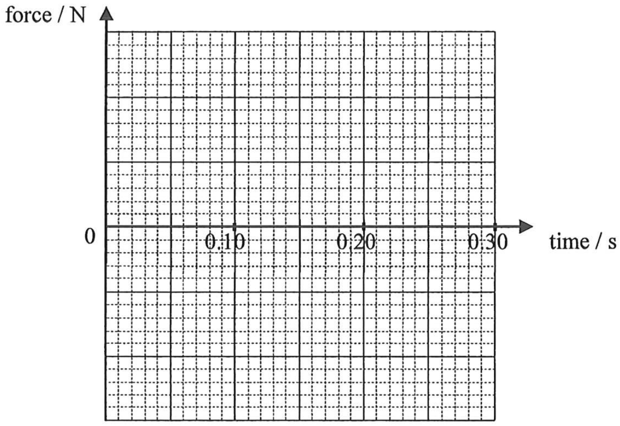
5. You are given two trolleys *A* and *B* (with mass of trolley *A* being larger), which can run freely on a smooth straight horizontal runway. Each trolley has a force sensor mounted on it and the two sensors are connected to a computer interface to record the time variation of force (force towards the right is registered as positive).

Figure 5.1



A student claims that when two objects collide, the less massive object experiences a larger force of collision. Describe how you would conduct an experiment to investigate the forces acting between the trolleys during collision. Sketch the expected force-time graphs of the trolleys in the grid provided and explain briefly. Assume that the collision takes place at time 0 s and lasts for approximately 0.1 s. (4 marks)



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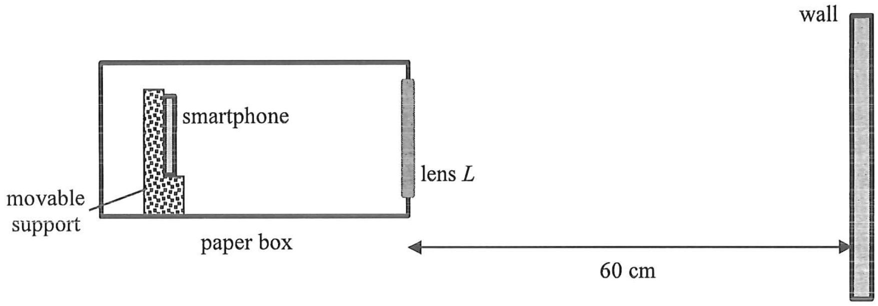
6. A student builds a smartphone projector as shown in Figure 6.1. It consists of a paper box with a lens L in front and a movable smartphone support inside. The position of the smartphone can be adjusted until a sharp image is formed on a wall.

Figure 6.1



(a) What kind of lens is used ? Explain briefly. (2 marks)

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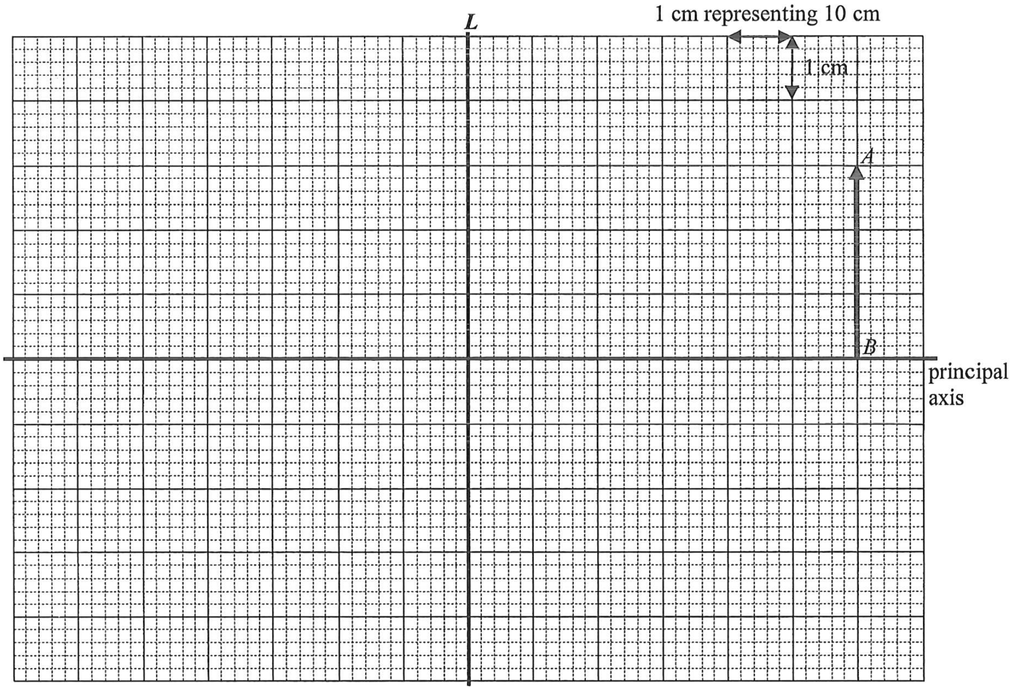
(b) In the figure below, AB represents the image formed on the wall, which is 60 cm from lens L. Given that the magnification of the image is 3.



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(i) Locate and draw the object (i.e. the display on the smartphone) using an arrow and denote it as O. (1 mark)

(ii) Complete the ray diagram by drawing **TWO** light rays reaching A from the object. (2 marks)

(iii) Hence, or otherwise, find the focal length of L. (1 mark)

(c) If the separation between lens L and the wall is slightly increased, state the adjustment to be made so as to restore a sharp image on the wall. (1 mark)

(d) Suggest a way to minimise unwanted reflections inside the box. (1 mark)

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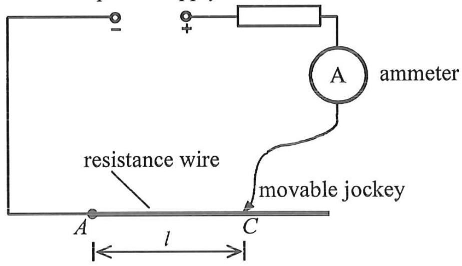
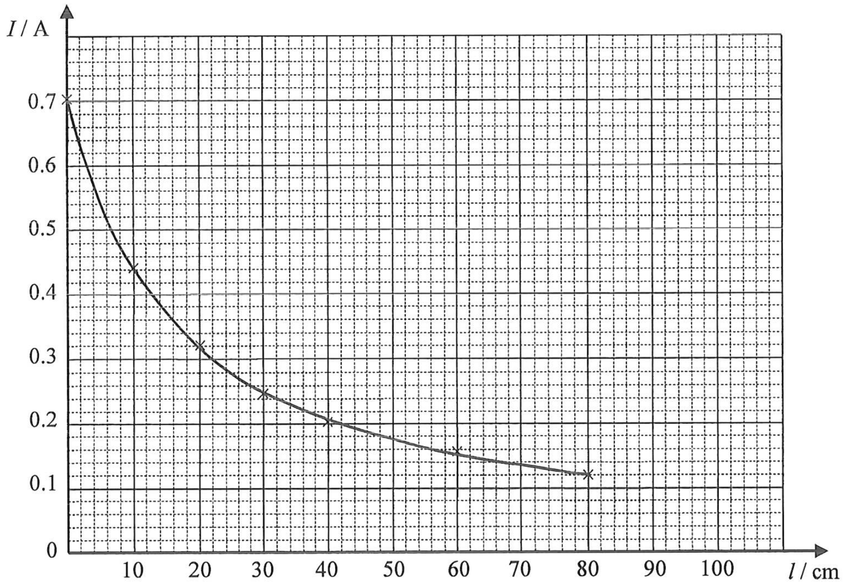
7. The circuit shown in Figure 7.1 can be used to measure the current at different lengths of a resistance wire using a movable jockey. The power supply of negligible internal resistance delivers a voltage of 3.5 V.

Figure 7.1



When the jockey is placed at point C on the resistance wire, the length of wire used is l and the corresponding current I is measured with an ammeter. The experiment is repeated at different lengths l and a graph of I against l is plotted below.



(a) What is the function of the 4.7 Ω resistor in this experiment ? (1 mark)

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(b) When the jockey is placed at point A (i.e. $l = 0$), find the resistance of the whole circuit. Compare it with the value of the 4.7 Ω resistor and account for the difference. (3 marks)

(c) When the jockey is moved from point A on the resistance wire to point C where $l = 30$ cm, find the increase in resistance of the circuit. (2 marks)

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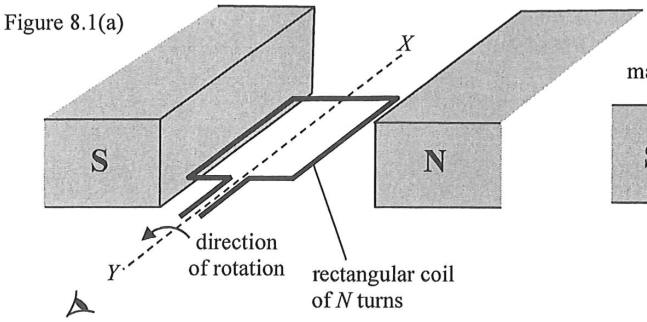
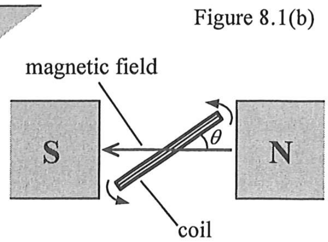
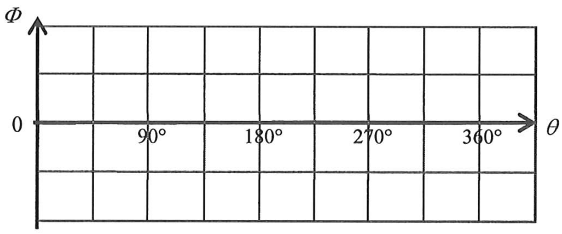
*8. A rectangular coil, with N turns and each turn of area 9.00×10^{-4} m^2, is placed in a uniform magnetic field of strength B. Initially the plane containing the coil is parallel to the magnetic field. The coil is then rotated steadily about the axis XY as shown in Figure 8.1(a).





(a) Sketch the variation of magnetic flux Φ passing through each turn of the coil when the coil rotates from $\theta = 0°$ to $360°$ (see Figure 8.1(b)). (2 marks)



(b) The maximum magnetic flux through each turn of the coil is 2.70×10^{-4} Wb.

(i) Determine B (unit: T). (2 marks)

(ii) Calculate the average induced e.m.f. ε in the coil when it is rotated from $\theta = 0°$ to $90°$ in 0.01 s. Given that the number of turns N is 100. (2 marks)

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9. An ideal gas is at temperature 0 °C and pressure 1.01×10^5 N m^{-2}.

(a) *(i) Find the volume of one mole of the gas. (2 marks)

(ii) Find the number of molecules in 1 m^3 of the gas under the conditions given. (2 marks)

*(b) At a certain location above the Earth in space where the pressure is 2.75×10^{-10} N m^{-2}, 1 m^3 of the gas contains about 1.00×10^{11} molecules. Assuming the gas is ideal, estimate the temperature at the location (in K). (2 marks)

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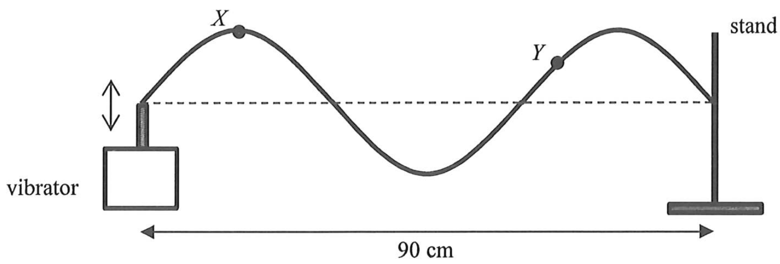
10. (a) A student uses the set-up in Figure 10.1 to investigate stationary waves on a stretched string. When the frequency of the vibrator is set at 100 Hz, a stationary wave is formed between the vibrator and the stand, which are 90 cm apart.

Figure 10.1



Particles X and Y on the string are at maximum displacement from their respective equilibrium positions at the moment shown in Figure 10.1.

(i) Label the positions of all antinodes (using 'A') and nodes (using 'N') on Figure 10.1. (2 marks)

(ii) Compare the oscillations of X and Y in terms of their amplitude and phase. (2 marks)

(iii) Sketch on Figure 10.1 the shape of the string just after 0.0075 s. (1 mark)

(iv) If the frequency of the vibrator is increased gradually, at what frequency will the next stationary wave pattern be observed ? Show your working. (2 marks)

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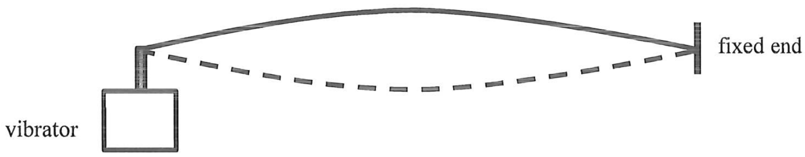
(b) Read the following passage about **stationary waves in musical instruments** and answer the questions that follow.

> How do we make musical sounds ? In non-electronic instruments such as guitar and piano, stable and controlled vibration is usually produced by a stationary wave. If you pluck a guitar string fixed at both ends, various modes of stationary waves will be produced and they all travel with the same speed on the string. The mode with the lowest frequency f_1 is called the fundamental while the n^{th} mode, called the n^{th} harmonic, has a frequency n times that of the fundamental, i.e. nf_1.
>
> The pitch of a note is determined by the frequency of the fundamental. The relative strengths and proportions of the harmonics determine the unique sound characteristics of different instruments. Therefore, a guitar and a violin playing a note of the same pitch will have different qualities because of different amplitudes and distributions of their harmonics.

(i) Figure 10.2 shows the stationary wave pattern of the fundamental formed on a string. Sketch the stationary wave pattern corresponding to the 2nd harmonic on Figure 10.2. (1 mark)

Figure 10.2



(ii) Explain why different musical instruments give different **sound qualities**. (2 marks)

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(iii) State **TWO** main differences between the **waves on a vibrating guitar string** and the **sound waves produced by it** besides the two waves being different in wavelength and speed. (2 marks)

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11. A coal-fired power plant consumes 4.5×10^6 kg of coal per day to generate electric power at an average rate of 680 MW. Given that burning 1 kg coal releases 32 MJ of energy.

(a) (i) Find the daily energy input (in MJ).

(2 marks)

(ii) Find the electrical energy produced per day (in MJ). (2 marks)

(iii) Estimate the efficiency of this power plant. (2 marks)

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(b) Transmission cables at 400 kV are used for long-distance electric power transmission to a city.

(i) Estimate the current in the cables. (2 marks)

(ii) Estimate the power loss (in kW) in the transmission cables. The resistance of the cables is $2.90 \times 10^{-8}\ \Omega$ per metre and the total length of the cables is 160 km. (2 marks)

*(iii) Explain why a.c. is preferred for electric power transmission. (2 marks)

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12. Carbon-14 ($^{14}_{6}C$) nuclide decays into stable nitrogen-14 ($^{14}_{7}N$) nuclide with a half-life of 5730 years. Assume that the concentration of carbon-14 in the atmosphere remains unchanged over time.

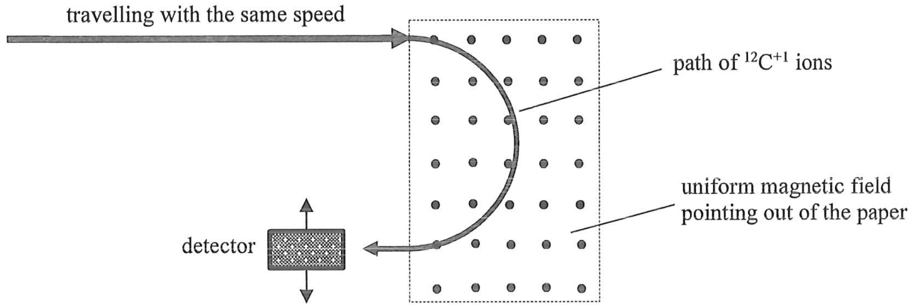
(a) (i) Write an equation to represent the decay of carbon-14. (1 mark)

*(ii) Find the decay constant (in yr^{-1}) of carbon-14. (2 marks)

*(iii) 1 g of an ancient wood sample has a carbon-14 activity that gives a count rate of 6.0 counts per minute. Estimate the age of this sample in years. Given that the carbon-14 activity in 1 g of fresh wood gives a count rate of 13.6 counts per minute. (2 marks)

(b) A more advanced method involves directly measuring the number of carbon-14 atoms relative to that of the carbon-12 atoms. Carbon atoms from the wood sample are ionized to form a beam which is directed into a magnetic field as shown in Figure 12.1. Each of the positive carbon-14 and carbon-12 ions possesses one electronic charge in magnitude.

Figure 12.1



(i) Both carbon-14 and carbon-12 are isotopes of carbon. What is meant by 'isotopes'? (1 mark)

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*(ii) Sketch the path of $^{14}C^{+1}$ ions on Figure 12.1. (1 mark)

(iii) Explain why the speed of the ions remains unchanged after passing through the magnetic field. (2 marks)

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END OF PAPER

Sources of materials used in this paper will be acknowledged in the *HKDSE Question Papers* booklet published by the Hong Kong Examinations and Assessment Authority at a later stage.

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